



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to HARUR APCMS BOMMIDI

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

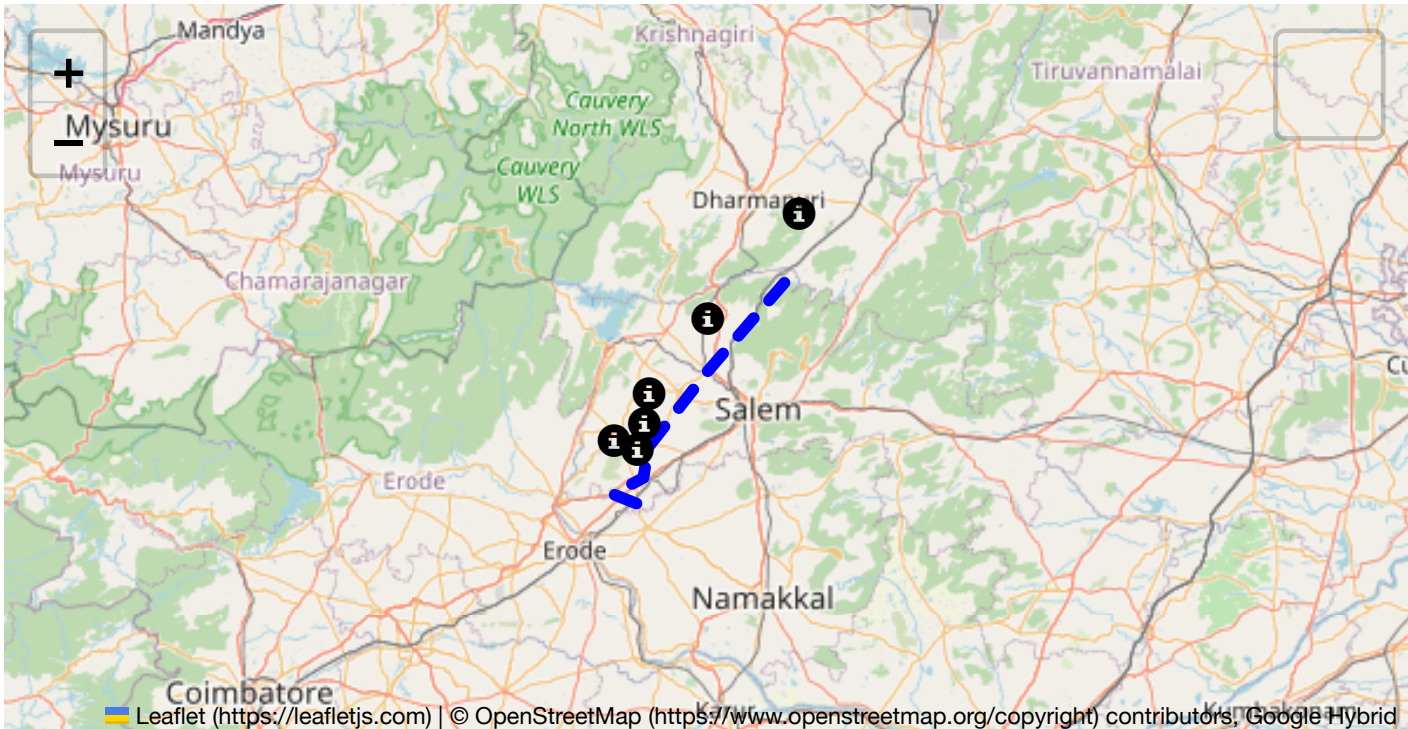
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

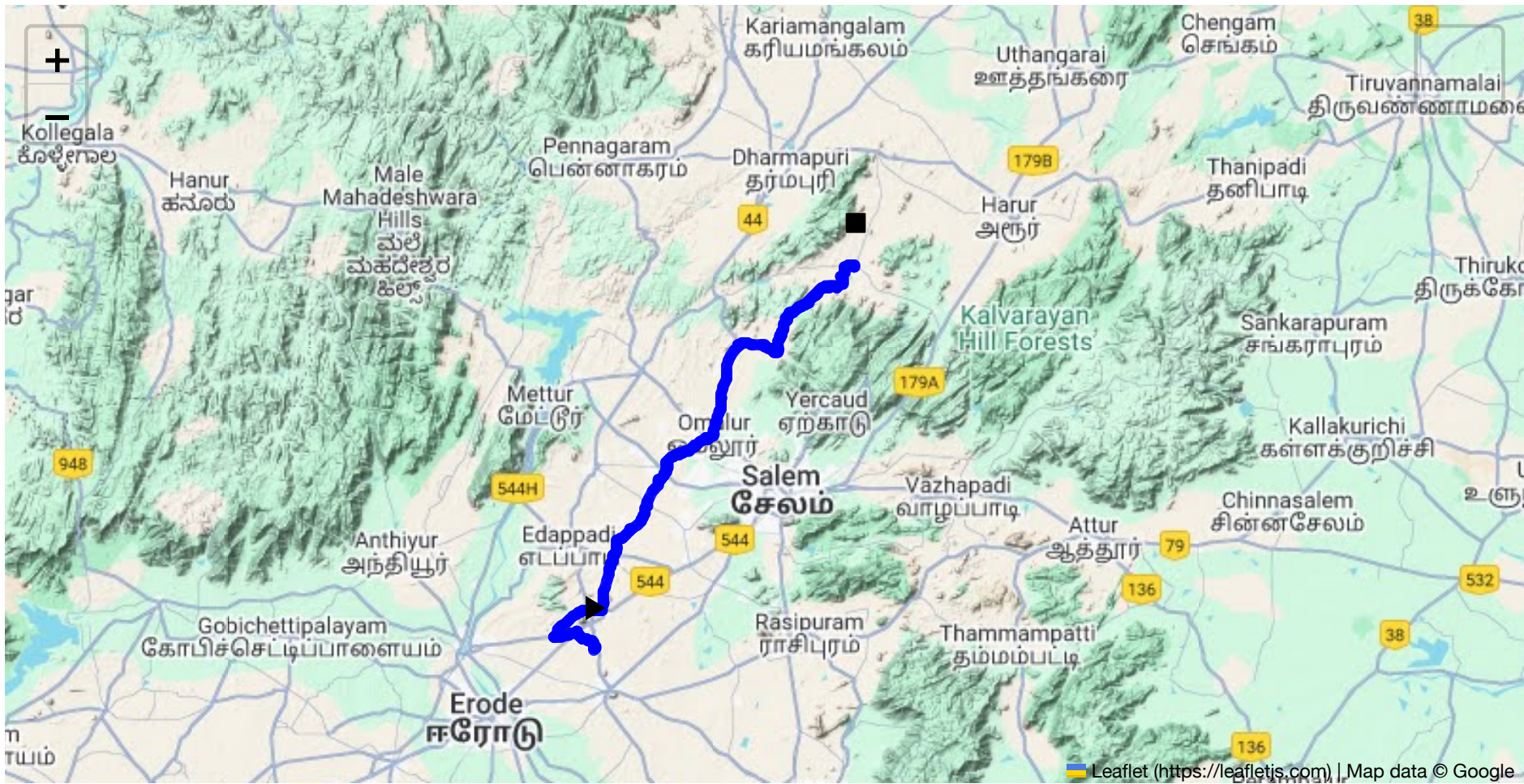
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 97.78 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours

Start: (11.4381, 77.8734)
End: (11.982729, 78.250069)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Bommidi spans about 97.78 kilometers and includes segments of Bhavani Main Road, SH 86, and passes through notable towns such as Padaiveedu, Ayveli, Erumaipatti, and Salem. The primary highways provide relatively direct connections with diversions through local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences tropical weather conditions. The route is susceptible to monsoon rains typically between June and September, which can lead to slippery roads and reduced visibility. During summer, heat can affect vehicle performance, especially in areas lacking shade or cooling facilities.

3. Analysis of Traffic Patterns

- Sangagiri to Salem:** Moderate traffic with potential congestion near urban centers.
- Salem:** Experiences heavy congestion, particularly during morning (8 AM - 10 AM) and evening (5 PM - 7 PM) peak hours.
- Salem to Bommidi:** Traffic generally flows smoothly, but expect slow movement through smaller towns.

4. Assessment of Road Quality and Infrastructure

- Bhavani Main Road & SH 86:** Generally well-maintained but may have sections under repair.
- Local Roads:** Vary in quality; some smaller connections may have poor surfaces and limited signage.

5. Suggestions for Alternative Routes for Emergencies

- In case of major incidents or closures on SH 86, consider NH 44 as an alternative for regional bypassing, although this might add considerable distance.
- Local diversions through smaller roads can provide temporary relief but may lack necessary support facilities.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu imposes regulations for the transportation of hazardous materials, including vehicle documentation, route approvals, and time restrictions. Ensure compliance with local diesel emission standards and proper handling certifications.

7. Overview of Historical Incidents

There have been occasional reports of accidents involving heavy vehicles in the Salem region, often linked to speeding and poor weather. Records specifically involving hazardous materials are sparse but underscore the importance of careful adherence to safety protocols.

8. Environmental Considerations and Sensitive Areas

- Several sections pass agricultural lands and water bodies, requiring caution to prevent contamination.
- Salem is an urban area with air quality management concerns, mandating reduced emissions.

9. Analysis of Communication Coverage

- Generally solid communication coverage in urban regions like Salem.
- Rural sections, particularly near Erumaipatti, might experience occasional network drops.

10. Estimated Emergency Response Times

- **Sangagiri to Salem:** Approximately 30-45 minutes.
- **Salem Urban Area:** 15-20 minutes.
- **Salem to Bommidi:** 45-60 minutes depending on proximity to smaller townships.

12. Overall Summary of Risk Assessment

The route presents a manageable risk level for hazardous material transport but requires attention to monsoon-related hazards, urban traffic in Salem, and compliance with local regulations. Coordination with local emergency services is advised to enhance response times in rural areas. Communication coverage is generally sufficient, with minor anticipated interruptions. Routine checks on vehicle and load conditions, adherence to safe speeds, and consideration of traffic patterns are critical to minimizing risks.

Key Recommendations: Ensure vehicles are well-equipped for varied weather, plan trips around peak congestion, maintain full compliance with transport regulations, and establish clear communication protocols for emergency situations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45853, 77.85139	30 KM/Hr
5	Turn	Medium	11.45906, 77.85132	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45636, 77.81678	15 KM/Hr
9	Turn	Medium	11.49383, 77.88501	30 KM/Hr
10	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
11	Turn	High	11.49469, 77.88488	15 KM/Hr
12	Turn	Medium	11.56072, 77.89919	30 KM/Hr
13	Turn	Medium	11.59085, 77.90592	30 KM/Hr
14	Turn	Medium	11.59253, 77.90690	30 KM/Hr
15	Turn	Medium	11.63848, 77.94825	30 KM/Hr
16	Turn	High	11.64197, 77.95197	15 KM/Hr
17	Turn	Medium	11.64558, 77.95108	30 KM/Hr
18	Turn	High	11.67870, 77.96846	15 KM/Hr
19	Turn	High	11.70227, 77.97864	15 KM/Hr
20	Turn	Medium	11.71386, 77.99552	30 KM/Hr
21	Turn	Medium	11.71735, 78.00019	30 KM/Hr
22	Turn	Medium	11.73785, 78.03496	30 KM/Hr
23	Turn	High	11.73653, 78.03886	15 KM/Hr
24	Turn	Medium	11.74252, 78.04705	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	High	11.87099, 78.08358	15 KM/Hr
28	Turn	High	11.87110, 78.08383	15 KM/Hr
29	Turn	Medium	11.87072, 78.08439	30 KM/Hr
30	Turn	Medium	11.87088, 78.08556	30 KM/Hr
31	Turn	High	11.87246, 78.08704	15 KM/Hr
32	Turn	Medium	11.86898, 78.12074	30 KM/Hr
33	Turn	Medium	11.86495, 78.12530	30 KM/Hr
34	Turn	High	11.85820, 78.13580	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	Medium	11.86044, 78.13937	30 KM/Hr
36	Turn	Medium	11.86652, 78.13879	30 KM/Hr
37	Turn	Medium	11.88411, 78.14478	30 KM/Hr
38	Turn	Medium	11.88540, 78.14474	30 KM/Hr
39	Turn	Medium	11.89899, 78.14930	30 KM/Hr
40	Turn	Medium	11.90581, 78.15292	30 KM/Hr
41	Turn	High	11.90760, 78.15205	15 KM/Hr
42	Turn	Medium	11.91226, 78.15594	30 KM/Hr
43	Turn	Medium	11.91312, 78.15616	30 KM/Hr
44	Turn	Medium	11.91519, 78.15837	30 KM/Hr
45	Turn	High	11.92454, 78.17073	15 KM/Hr
46	Turn	Medium	11.92448, 78.17340	30 KM/Hr
47	Turn	Medium	11.92820, 78.18040	30 KM/Hr
48	Turn	Medium	11.92797, 78.18165	30 KM/Hr
49	Turn	Medium	11.93585, 78.18730	30 KM/Hr
50	Turn	Medium	11.95116, 78.20647	30 KM/Hr
51	Turn	Medium	11.95200, 78.20703	30 KM/Hr
52	Turn	High	11.95258, 78.20848	15 KM/Hr
53	Turn	Medium	11.95173, 78.21017	30 KM/Hr
54	Turn	High	11.95221, 78.21173	15 KM/Hr
55	Turn	Medium	11.95165, 78.21276	30 KM/Hr
56	Turn	Medium	11.95233, 78.21464	30 KM/Hr
57	Turn	High	11.95220, 78.21522	15 KM/Hr
58	Turn	Medium	11.95288, 78.21603	30 KM/Hr
59	Turn	Medium	11.95264, 78.22467	30 KM/Hr
60	Turn	High	11.95178, 78.22600	15 KM/Hr
61	Turn	Medium	11.95847, 78.23410	30 KM/Hr
62	Turn	Medium	11.96509, 78.23292	30 KM/Hr
63	Turn	High	11.96652, 78.23302	15 KM/Hr
64	Blind Spot	Blind Spot	11.96679, 78.23211	10 KM/Hr
65	Turn	High	11.97329, 78.23713	15 KM/Hr
66	Turn	High	11.97765, 78.23556	15 KM/Hr
67	Turn	High	11.98240, 78.24290	15 KM/Hr
68	Turn	High	11.98205, 78.24333	15 KM/Hr
69	Turn	High	11.98346, 78.24637	15 KM/Hr
70	Blind Spot	Blind Spot	11.98114, 78.24921	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
30	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
31	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
32	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
33	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
34	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
35	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium
36	hospital	Raja Hospital, Dharmapuri	11.9792908, 78.2509482	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87099, 78.08358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87110, 78.08383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87072, 78.08439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87088, 78.08556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87246, 78.08704



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86898, 78.12074



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95116, 78.20647



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97329, 78.23713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97765, 78.23556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98240, 78.24290



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98205, 78.24333



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98346, 78.24637



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.98114, 78.24921

